
Problem Statement Title: AI-Powered Finance Education & Investment Intelligence Platform

Problem Statement:

Despite increased access to financial markets, there remains a gap between financial learning and real-world application. Many individuals understand theoretical concepts but struggle to interpret market movements, analyze news impact, or manage portfolios effectively.

Existing platforms separate learning, trading tools, and advisory systems. There is a lack of an integrated ecosystem that allows users to:

- Learn financial concepts
 - Practice market prediction
 - Understand the effect of news on markets
 - Analyze personal portfolios
 - Receive AI-driven strategic guidance
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Challenge:

Build a full-stack web platform that integrates financial education with real-time market intelligence and AI-powered advisory.

The platform should function as a unified ecosystem where users can learn, practice, analyze, and receive intelligent guidance within a single environment.

The Platform Should Feature:

- **User Authentication & Profiles:** Secure login system allowing users to track learning progress, prediction history, and portfolio insights.
- **Finance Learning & Assessment Module:** A structured learning environment with assessments and progress tracking.

- **Stock Prediction Playground:** An interactive and appealing paper-trading arena where users predict stock movement and compare their predictions with the platform's AI-generated prediction (ML) and actual market outcome. The system must generate intelligent explanations for outcomes, regardless of whether predictions match or differ.
 - **Financial News Intelligence Module:** A live news analysis section that evaluates financial news (ML) and visually demonstrates its impact on market movement.
 - **Portfolio Analyzer:** A portfolio analysis dashboard that provides insights into allocation, performance, and risk exposure.
 - **AI Financial Strategy Advisor:** A **conversational advisor chatbot** that uses insights from the News Module and Portfolio Analyzer, Stock Prediction Playground to provide personalized strategy suggestions and should answer any finance related queries of the user.
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Bonus Points For (+5 for each):

- **Web3 Authentication:** Decentralized login using crypto wallet integration.
 - **Live Deployment:** The platform must be publicly accessible with a working demo.
 - **Enhanced Interactive Visualizations:** Advanced visual analytics to demonstrate market-news relationships, portfolio insights and stock market playground.
 - **Community Build Up:** A working community chat for users and like-minded people to share insights on trading and finance.
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Objective:

Create a scalable, user-friendly, and intelligent full-stack financial ecosystem that bridges education, real-world market understanding, and AI-driven advisory into a single platform.

The solution should simplify financial intelligence while preserving analytical depth.

Optional Deliverables:

- **Postman Documentation for API testing**

General Rules:

- All code must be developed during the hackathon timeframe.
 - Final submissions must be uploaded to GitHub before the deadline.
 - Teams must present a working live demo to the judges.
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Development Guidelines:

- Tech Stack: Any modern stack is allowed.
 - Architecture: Backend and frontend must be clearly separated.
 - Security: Follow best practices for authentication and API protection.
 - DevOps: Use Git with meaningful commits.
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Evaluation Criteria:

Criteria	Marks	Focus Point
System Architecture	25	Clarity, modularity, and structured system design
AI Integration & Functional Logic	25	Intelligent prediction, analysis, and advisory flow
Full-Stack Implementation	20	End-to-end flow and module integration
Visualization & User Experience	15	Clarity of insights and intuitive design
Innovation & Creativity	15	Unique features and strategic thinking

Total: 100 Marks